

§ 33.49 Endurance test. (a) *General.* Each engine must be subjected to an endurance test that includes a total of 150 hours of operation (except as provided in paragraph (e)(1)(iii) of this section) and, depending upon the type and contemplated use of the engine, consists of one of the series of runs specified in paragraphs (b) through (e) of this section, as applicable. The runs must be made in the order found appropriate by the Administrator for the particular engine being tested. During the endurance test the engine power and the crankshaft rotational speed must be kept within ± 3 percent of the rated values. During the runs at rated take-off power and for at least 35 hours at rated maximum continuous power, one cylinder must be operated at not less than the limiting temperature, the other cylinders must be operated at a temperature not lower than 50 degrees F. below the limiting temperature, and the oil inlet temperature must be maintained within ± 10 degrees F. of the limiting temperature. An engine that is equipped with a propeller shaft must be fitted for the endurance test with a propeller that thrust-loads the engine to the maximum thrust which the engine is designed to resist at each applicable operating condition specified in this section. Each accessory drive and mounting attachment must be loaded. During operation at rated takeoff power and rated maximum continuous power, the load imposed by each accessory used only for an aircraft service must be the limit load specified by the applicant for the engine drive or attachment point. (b) *Unsupercharged engines and engines incorporating a gear-driven single-speed supercharger.* For engines not incorporating a supercharger and for engines incorporating a gear-driven single-speed supercharger the applicant must conduct the following runs: (1) A 30-hour run consisting of alternate periods of 5 minutes at rated -takeoff power with takeoff speed, and 5 minutes at maximum best economy cruising power or maximum recommended cruising power. (2) A 20-hour run consisting of alternate periods of 1 1/2 hours at rated maximum continuous power with maximum continuous speed, and 1/2 hour at 75 percent rated maximum continuous power and 91 percent maximum continuous speed. (3) A 20-hour run consisting of alternate periods of 1 1/2 hours at rated maximum continuous power with maximum continuous speed, and 1/2 hour at 70 percent rated maximum continuous power and 89 percent maximum continuous speed. (4) A 20-hour run consisting of alternate periods of 1 1/2 hours at rated maximum continuous power with maximum continuous speed, and 1/2 hour at 65 percent rated maximum continuous power and 87 percent maximum continuous speed. (5) A 20-hour run consisting of alternate periods of 1 1/2 hours at rated maximum continuous power with maximum continuous speed, and 1/2 hour at 60 percent rated maximum continuous power and 84.5 percent maximum continuous speed. (6) A 20-hour run consisting of alternate periods of 1 1/2 hours at rated maximum continuous power with maximum continuous speed, and 1/2 hour at 50 percent rated maximum continuous power and 79.5 percent maximum continuous speed. (7) A 20-hour run consisting of alternate periods of 2 1/2 hours at rated maximum continuous power with maximum continuous speed, and 2 1/2 hours at maximum best economy cruising power or at maximum recommended cruising power. (c) *Engines incorporating a gear-driven two-speed supercharger.* For engines incorporating a gear-driven two-speed supercharger the applicant must conduct the following runs: (1) A 30-hour run consisting of alternate periods in the lower gear ratio of 5 minutes at rated takeoff power with takeoff speed, and 5 minutes at maximum best economy cruising power or at maximum recommended cruising power. If a takeoff power rating is desired in the higher gear ratio, 15 hours of the 30-hour run must be made in the higher gear ratio in alternate periods of 5 minutes at the observed horsepower obtainable with the -takeoff critical altitude manifold pressure and takeoff speed, and 5 minutes at 70 percent high ratio rated maximum continuous power and 89 percent high ratio maximum continuous speed. (2) A 15-hour run consisting of alternate periods in the lower gear ratio of 1 hour at rated maximum continuous power with maximum continuous speed, and 1/2 hour at 75 percent rated maximum continuous power and 91 percent maximum continuous speed. (3) A 15-hour run consisting of alternate periods in the lower gear ratio of 1 hour at rated maximum continuous power with maximum continuous speed, and 1/2 hour at 70 percent rated maximum continuous power and 89 percent maximum continuous speed. (4) A 30-hour run in the higher gear ratio at rated maximum continuous power with maximum continuous speed. (5) A 5-hour run consisting of alternate periods of 5 minutes in each of the supercharger gear ratios. The first 5 minutes of the test must be made at maximum continuous speed in the higher gear ratio and the observed horsepower obtainable with 90 percent of maximum continuous manifold pressure in the higher gear ratio under sea level conditions. The condition for operation for the alternate 5 minutes in the lower gear ratio must be that obtained by shifting to the lower gear ratio at constant speed. (6) A 10-hour run consisting of alternate periods in the lower gear ratio of 1 hour at rated maximum continuous power with maximum continuous speed, and 1 hour at 65 percent rated maximum continuous power and 87 percent maximum continuous

speed. (7) A 10-hour run consisting of alternate periods in the lower gear ratio of 1 hour at rated maximum continuous power with maximum continuous speed, and 1 hour at 60 percent rated maximum continuous power and 84.5 percent maximum continuous speed. (8) A 10-hour run consisting of alternate periods in the lower gear ratio of 1 hour at rated maximum continuous power with maximum continuous speed, and 1 hour at 50 percent rated maximum continuous power and 79.5 percent maximum continuous speed. (9) A 20-hour run consisting of alternate periods in the lower gear ratio of 2 hours at rated maximum continuous power with maximum continuous speed, and 2 hours at maximum best economy cruising power and speed or at maximum recommended cruising power. (10) A 5-hour run in the lower gear ratio at maximum best economy cruising power and speed or at maximum recommended cruising power and speed. Where simulated altitude test equipment is not available when operating in the higher gear ratio, the runs may be made at the observed horsepower obtained with the critical altitude manifold pressure or specified percentages thereof, and the fuel-air mixtures may be adjusted to be rich enough to suppress detonation. (d) *Helicopter engines.* To be eligible for use on a helicopter each engine must either comply with paragraphs (a) through (j) of § 29.923 of this chapter, or must undergo the following series of runs: (1) A 35-hour run consisting of alternate periods of 30 minutes each at rated takeoff power with takeoff speed, and at rated maximum continuous power with maximum continuous speed. (2) A 25-hour run consisting of alternate periods of 2½ hours each at rated maximum continuous power with maximum continuous speed, and at 70 percent rated maximum continuous power with maximum continuous speed. (3) A 25-hour run consisting of alternate periods of 2½ hours each at rated maximum continuous power with maximum continuous speed, and at 70 percent rated maximum continuous power with 80 to 90 percent maximum continuous speed. (4) A 25-hour run consisting of alternate periods of 2½ hours each at 30 percent rated maximum continuous power with takeoff speed, and at 30 percent rated maximum continuous power with 80 to 90 percent maximum continuous speed. (5) A 25-hour run consisting of alternate periods of 2½ hours each at 80 percent rated maximum continuous power with takeoff speed, and at either rated maximum continuous power with 110 percent maximum continuous speed or at rated takeoff power with 103 percent takeoff speed, whichever results in the greater speed. (6) A 15-hour run at 105 percent rated maximum continuous power with 105 percent maximum continuous speed or at full throttle and corresponding speed at standard sea level carburetor entrance pressure, if 105 percent of the rated maximum continuous power is not exceeded. (e) *Turbosupercharged engines.* For engines incorporating a turbosupercharger the following apply except that altitude testing may be simulated provided the applicant shows that the engine and supercharger are being subjected to mechanical loads and operating temperatures no less severe than if run at actual altitude conditions: (1) For engines used in airplanes the applicant must conduct the runs specified in paragraph (b) of this section, except— (i) The entire run specified in paragraph (b)(1) of this section must be made at sea level altitude pressure; (ii) The portions of the runs specified in paragraphs (b)(2) through (7) of this section at rated maximum continuous power must be made at critical altitude pressure, and the portions of the runs at other power must be made at 8,000 feet altitude pressure; and (iii) The turbosupercharger used during the 150-hour endurance test must be run on the bench for an additional 50 hours at the limiting turbine wheel inlet gas temperature and rotational speed for rated maximum continuous power operation unless the limiting temperature and speed are maintained during 50 hours of the rated maximum continuous power operation. (2) For engines used in helicopters the applicant must conduct the runs specified in paragraph (d) of this section, except— (i) The entire run specified in paragraph (d)(1) of this section must be made at critical altitude pressure; (ii) The portions of the runs specified in paragraphs (d)(2) and (3) of this section at rated maximum continuous power must be made at critical altitude pressure and the portions of the runs at other power must be made at 8,000 feet altitude pressure; (iii) The entire run specified in paragraph (d)(4) of this section must be made at 8,000 feet altitude pressure; (iv) The portion of the runs specified in paragraph (d)(5) of this section at 80 percent of rated maximum continuous power must be made at 8,000 feet altitude pressure and the portions of the runs at other power must be made at critical altitude pressure; (v) The entire run specified in paragraph (d)(6) of this section must be made at critical altitude pressure; and (vi) The turbosupercharger used during the endurance test must be run on the bench for 50 hours at the limiting turbine wheel inlet gas temperature and rotational speed for rated maximum continuous power operation unless the limiting temperature and speed are maintained during 50 hours of the rated maximum continuous power operation. [Amdt. 33-3, 32 FR 3736, Mar. 4, 1967, as amended by Amdt. 33-6, 39 FR 35465, Oct. 1, 1974; Amdt. 33-10, 49 FR 6851, Feb. 23, 1984]

Sinngemäße deutsche Übersetzung nach bestem Wissen:

§ 33.49 Dauerprüfung: Allgemeines. Jeder Motor muss einer Dauerprüfung unterzogen werden, die insgesamt 150 Betriebsstunden umfasst (mit Ausnahme der Bestimmungen in Absatz (e) (1) (iii) dieses Abschnitts). Je nach Art und beabsichtigter Verwendung des Motors besteht die Dauerprüfung aus Läufen mit vorgegebenem Zyklus. Diese Zyklen sind in den Abschnitten (b) bis (e) dieses Abschnitts angegeben. Die Läufe müssen in der Reihenfolge durchgeführt werden, wie sie vom Verantwortlichen für den zu testenden Motor vorgegeben werden.

Während der Dauerprüfung darf die Motorleistung und die Drehzahl der Kurbelwelle maximal $\pm 3\%$ von den Nennwerten abweichen.

Während der Läufe bei Startleistung (take-off power) und für die letzten 35 Stunden bei maximaler Dauerleistung muss mindestens ein Zylinder an der Grenztemperatur betrieben werden. Die übrigen Zylinder dürfen maximal 10°C (50°F) unterhalb der Grenztemperatur betrieben werden. Die Öltemperatur muss dabei an der Grenztemperatur betrieben werden und darf maximal $\pm 10^{\circ}\text{F}$ (ca. $5,5^{\circ}\text{C}$) abweichen. Ein Motor an dem direkt ein Propeller angebracht werden kann, muss für die Dauerprüfung mit einem Propeller ausgestattet sein, der die beschriebenen Betriebszustände hervorrufen kann und den Motor damit maximal belastet.

Alle Anbauteile und Hilfsaggregate müssen angebracht sein. Während des Betriebs bei Startleistung (take-off power) und maximaler Dauerleistung müssen die Hilfsaggregate oder Zubehörteile auch mit Maximalleistung betrieben werden.

(b) Motoren ohne Turbolader oder Kompressor. Für Motoren ohne Turbolader oder Kompressor müssen Läufe mit folgendem Zyklus durchgeführt werden:

1. 30-Stunden-Lauf: Im 5 Minuten Takt wird zwischen Startleistung/Startdrehzahl (take-off power) und dem Betriebspunkt bei Reiseflug bzw. dem Wirtschaftlichsten Betriebspunkt gewechselt.
2. 20-Stunden-Lauf: Der Motor wird abwechselnd 1,5 Stunden bei maximaler Leistung und Drehzahl und 0,5 Stunden bei 75% der Leistung und 91% der Drehzahl betrieben.
3. 20-Stunden-Lauf: Der Motor wird abwechselnd 1,5 Stunden bei maximaler Leistung und Drehzahl und 0,5 Stunden bei 70% der Leistung und 89% der Drehzahl betrieben.
4. 20-Stunden-Lauf: Der Motor wird abwechselnd 1,5 Stunden bei maximaler Leistung und Drehzahl und 0,5 Stunden bei 65% der Leistung und 87% der Drehzahl betrieben.
5. 20-Stunden-Lauf: Der Motor wird abwechselnd 1,5 Stunden bei maximaler Leistung und Drehzahl und 0,5 Stunden bei 60% der Leistung und 84,5% der Drehzahl betrieben.
6. 20-Stunden-Lauf: Der Motor wird abwechselnd 1,5 Stunden bei maximaler Leistung und Drehzahl und 0,5 Stunden bei 50% der Leistung und 79,5% der Drehzahl betrieben.
7. 20-Stunden-Lauf: Der Motor wird abwechselnd 2,5 Stunden bei maximaler Leistung und Drehzahl und 2,5 Stunden am Betriebspunkt bei Reiseflug bzw. dem Wirtschaftlichsten Betriebspunkt betrieben.